EXERCISE 14

Non-Lipschitz Branching and Picard's Theorem

Original statement · ProblemsCh2.pdf, p. 4

Non-uniqueness • Failure of Lipschitz

PROBLEM STATEMENT

Given the IVP

$$\frac{dy}{dt} = 3y^{2/3}, \qquad y(0) = 0,$$

on the rectangle $R = \{(y,t) \in \mathbb{R}^2 : |y| \le 1,\ |t| \le 1\}$. Show that both $y_1(t) = t^3$ and $y_2(t) = 0$ are solutions. Does this contradict Picard's theorem?

Source: ProblemsCh2.pdf, Exercise 14 (page 4).

PHASE 1: CLASSIFICATION, HYPOTHESES AND DOMAIN

Right-hand side $f(y,t) = 3y^{2/3}$, where $y^{2/3} = (y^{1/3})^2 = |y|^{2/3} \ge 0$ is defined for all real y (real cube root). It is continuous on R. Its partial derivative is

$$\frac{\partial f}{\partial y} = 2\,y^{-1/3} \quad (y \neq 0),$$

which is unbounded as $y \to 0$ and not defined at $y = 0$, which is precisely the initial value $y(0) = 0$. So the hypotheses of Picard's theorem (continuity of f and $\partial f/\partial y$ near the initial point, or a Lipschitz condition in y) are not satisfied at $(0,0)$.

PHASE 2: CHOICE OF METHOD AND CHANGE OF VARIABLES

Why this method: to show that a candidate is a solution of an IVP we verify the ODE and the initial condition by direct substitution. To decide whether the non-uniqueness contradicts Picard's theorem, we test the Lipschitz condition $|f(y,t) - f(\tilde{y},t)| \le L|y-\tilde{y}|$ near $y=0$; if it fails, the theorem does not apply and no contradiction arises.

METHOD OF THE CLASS NOTES [NOTES, P. 3]

In the form of the notes, $y' = f(t,y)$ with $f = 3y^{2/3}$, Picard's theorem needs f and $\partial f/\partial y$ continuous near $(t_0,y_0) = (0,0)$. Here f is continuous, but $\frac{\partial f}{\partial y} = 3 \cdot \frac{2}{3}\, y^{-1/3} = 2y^{-1/3}$ is not defined at $y=0$ and is unbounded as $y \to 0$. The hypothesis fails, so the theorem says nothing at $(0,0)$; the Lipschitz test below (Robinson's form of the hypothesis) reaches the same conclusion.

PHASE 3: STEP-BY-STEP DERIVATION

Verification of $y_1(t) = t^3$

Initial condition: $y_1(0) = 0$. Derivative: $y_1'(t) = 3t^2$. Right-hand side with the real cube root: $y_1^{1/3} = (t^3)^{1/3} = t$, so $y_1^{2/3} = t^2$ and $3y_1^{2/3} = 3t^2$. Both sides agree for all t. The graph stays in R for $|t| \le 1$, since $|y_1| = |t|^3 \le 1$.

Verification of $y_2(t) = 0$

$y_2' = 0$ and $3 \cdot 0^{2/3} = 0$, and $y_2(0) = 0$. It is a solution, and its graph is inside R.

Two distinct solutions, $y_1(t) = t^3$ and $y_2(t) = 0$, of the same IVP exist on $[-1,1]$.

Failure of the Lipschitz condition

Take $\tilde{y} = 0$ and $y > 0$:

$$\frac{|f(y,t) - f(0,t)|}{|y-0|} = \frac{3y^{2/3}}{y} = 3\,y^{-1/3} \longrightarrow +\infty \quad (y \to 0^+).$$

No constant L bounds this quotient, so f is not Lipschitz in y on any neighborhood of $y=0$ in R. Consistently, $\partial f/\partial y = 2y^{-1/3}$ is unbounded.

Where the branching comes from

For $y > 0$ the equation is separable: $\dfrac{dy}{3y^{2/3}} = dt$. Integrating, $y^{1/3} = t + c$, so $y = (t+c)^3$ for $t + c > 0$. For every $c \in [0,1]$ define

$$y_c(t) = \begin{cases} 0, & -1 \le t \le c, \\ (t-c)^3, & c < t \le 1. \end{cases}$$

At $t = c$ the one-sided derivatives are both 0 ($\frac{d}{dt}(t-c)^3 = 3(t-c)^2 \to 0$), so y_c is C^1, it satisfies the ODE on each piece and $y_c(0) = 0$. Hence the IVP has infinitely many solutions, and $y_2 = 0$ (the limit $c \ge 1$) is a member of the family. Note that y_c with $c = 0$ is 0 for $t \le 0$ and t^3 for $t > 0$; the function $y_1 = t^3$ on all of $[-1, 1]$ is a different solution, obtained by using the negative-side branch $y = (t - c)^3, t < c$, with $c = 0$ (valid since $y^{1/3} = t - c$ requires only y and $t - c$ to have the same sign, and $\frac{d}{dt}t^3 = 3t^2 = 3(t^3)^{2/3}$).

PHASE 4: VERIFICATION, LIMITS AND INTERPRETATION

Answer to the question: no contradiction. Picard's theorem (Robinson, Theorem 6.2) guarantees uniqueness only if f and $\partial f/\partial y$ are continuous (or f is Lipschitz in y) near the initial point. Here f is continuous but $\partial f/\partial y = 2y^{-1/3}$ is not defined at $y = 0$ and the Lipschitz quotient is unbounded, so the hypotheses fail exactly at the initial point $(0, 0)$. The theorem is silent in that case, and non-uniqueness is possible.

Contrast: for any initial value $y_0 \neq 0$ with $t_0 \in (-1, 1)$, f and $\partial f/\partial y$ are continuous in a neighborhood and the solution is unique; the branching occurs only from the equilibrium $y = 0$ (compare Problem 2.9 (iii) and (iv)).

Why the solution can leave $y = 0$: the time to travel from 0 to y is $\int_0^y ds/(3s^{2/3}) = y^{1/3}$, which is finite and tends to 0 as $y \to 0$. For Lipschitz right-hand sides this integral diverges (compare $f = y$, where $\int_0 ds/s = \infty$), so an equilibrium cannot be left in finite time.

RESULT

$y_1 = t^3$ and $y_2 = 0$ both solve $\dot{y} = 3y^{2/3}, y(0) = 0$. No contradiction with Picard's theorem: its hypotheses fail at $(0, 0)$ because $\partial f/\partial y = 2y^{-1/3}$ is unbounded.